

Response to Reviewers

Trading-Aware Movement in Sugarscape: A Deep Reinforcement Learning Approach to Adaptive Economic Behavior

Computational Economics — Revision 1 (Major Revision)

We thank both reviewers for their careful and constructive reading of the manuscript. The comments identified a genuine mismatch between the paper’s original framing and its actual method, along with several points where the technical presentation and empirical claims needed tightening. We have revised the manuscript substantially in response. The central change is a reframing of the contribution around **trade-aware movement**—a learned movement policy conditioned on local market information, with bilateral exchange executed by the classical Rule T—which resolves the framing concerns raised by Reviewer 1 and clarifies the scope of what is and is not learned.

Below we respond to each point in turn. We have chosen throughout to state openly what was changed and where, and to be candid about the precise scope of each new result rather than to overstate what was done. Section references are to the revised manuscript.

Response to Reviewer #1

R1.1 — Framing overstates joint optimization of movement and exchange

“The framing repeatedly suggests ‘learning trading strategies’ and ‘jointly optimizing movement and exchange,’ but the algorithmic design is movement-only learning + exogenous trading... a major mismatch between rhetoric and method.”

The reviewer is correct, and this was the root cause underlying several of the points below. The manuscript has been reframed throughout around *trade-aware movement*: the learned policy governs movement only, while bilateral exchange continues to be executed by the classical Rule T. Every passage that previously claimed joint optimization has been rewritten—the abstract, the introduction, and Sections 2.1–2.4. A new **“Scope of Learning”** paragraph now opens Section 3.2, stating explicitly that the policy has no control over what, how much, at what price, or whether to trade, and that the term “trade-aware” refers to the conditioning of movement on neighbours’ MRS. A full-document check confirms no remaining instances of “jointly optimize,” “unifies movement and trade,” or “learns trading strategies.”

R1.2 — The true differentiator is movement learning conditioned on trading-related state

“The differentiator is not ‘RL in Sugarscape’ but ‘movement learning conditioned on trading-related state while trade remains rule-based.’”

We have adopted precisely this characterization as the paper’s stated contribution. The introduction now presents a threefold contribution: (i) a conceptual update to Sugarscape that embeds bounded-rational adaptation *without* replacing the classical trading mechanism; (ii) a methodologically controlled comparison between Rule-M and PPO-learned movement under matched conditions; and (iii) a computational laboratory for policy experiments. Section 2.2 repositions the novelty relative to prior RL-in-Sugarscape work (Brearcliffe & Crooks; Jäger) exactly as the

reviewer frames it: those works learned movement over resource state alone, whereas the present work embeds *market information* (neighbours’ MRS) into the observation so that movement can respond to trading opportunity.

R1.3 — Baseline and DRL regimes differ in multiple parameters, invalidating causal attribution

“The baseline and deep RL regimes differ in multiple structural parameters simultaneously, making causal attribution to RL invalid.”

This was a presentation problem rather than a design problem, and we have corrected the presentation so that the point of confusion is removed. The original two-column table juxtaposed the *canonical Epstein–Artell 1996 configuration* (shown only for reference) against *this study’s settings*, which made it appear that the rule-based and DRL conditions used different parameters. In fact, both conditions in every experiment share identical environmental parameters and differ only in the movement policy. We have (a) replaced the two-column table with a single-column table of this study’s parameters, whose caption states that all values apply identically to both conditions; (b) added an explicit identity-of-conditions statement in Section 3.1 listing every held-fixed parameter; and (c) echoed that statement in Section 4.1 immediately before the headline carrying-capacity result, so that causal attribution to the movement policy is established at the point of the claim.

R1.4 — “Agents learn trading strategies” contradicts “trading is not an explicit action”

“These cannot both be true in the usual RL sense. . . the ‘trading-aware’ claim is inconsistent with the action design.”

Resolved by the same reframe as R1.1. The specific sentence the reviewer quoted (Section 2.2) has been rewritten so that agents learn *movement* strategies conditioned on neighbours’ MRS, positioning themselves for exchanges executed by Rule T. Section 2.3’s claim that the action space includes “implicit trading decisions” has been corrected to state that the action space comprises directional moves only. The Scope of Learning paragraph makes the two statements consistent: trade is not a learned action, and “trade-aware” refers strictly to the observation and the movement response to it.

R1.5 — “Ensures the Markov property” is a misconception in multi-agent settings

“From the perspective of one agent, the environment is generally non-stationary. . . the joint dynamics are not a standard single-agent MDP.”

Section 3.3.1 has been rewritten and retitled “Decision Problem and Decentralized Learning.” The joint system is now correctly identified as a homogeneous decentralized partially observable Markov decision process (Dec-POMDP) following Oliehoek & Amato (2016). Non-stationarity arising from co-evolving policies is stated explicitly, citing Lowe et al. (2017) and Foerster et al. (2018); parameter sharing is presented as a partial mitigation rather than a resolution, and the training regime is described precisely as PPO with full parameter sharing and decentralized execution—a single shared policy trained on the pooled experience of all agents, each acting on its own local observation (Yu et al. 2022). The phrase “ensures the Markov property” has been replaced by the precise claim that the observation is Markovian *for the decentralized policy*, with

its meaning defined.

R1.6 — The observation equation is unclear and possibly dimensionally incorrect

“The equation... is unclear and potentially dimensionally incorrect.”

Section 3.2.1 now specifies the observation fully. The agent observes the four cardinal arms of its von Neumann neighbourhood (v cells in each of N/S/E/W plus its own cell), giving $n = 4v + 1 = 41$ cells at $v = 10$ —a “cross-shaped” design that deliberately mirrors the directional action space, with unreachable diagonal cells excluded. The four channels per cell are enumerated (the welfare-surplus channel given as an explicit formula), and the flattening and concatenation with the six-element internal vector to a **170-dimensional** input is stated explicitly, removing all dimensional ambiguity.

R1.7 — Reward definition is underspecified and possibly inconsistent

“Reward definition is underspecified and may be conceptually inconsistent, please tighten it.”

Section 3.2.3 now gives the reward as a numbered equation and commits to a single, consistent interpretation. The reward is the one-step change in Cobb–Douglas welfare between the current state and the candidate destination under a *deterministic forward model* of movement and metabolism, minus a death penalty. We name this an *expected-utility heuristic* and state its exact scope: it is exact for the deterministic components and may diverge from realized welfare only through stochastic within-step events. We relate it to potential-based reward shaping (Ng et al. 1999), including the honest note that formal potential-based shaping would carry a discount factor our welfare-difference form omits (negligible at $\gamma = 0.99$). The death penalty is defined as a configurable hyperparameter set to 1 in all experiments. Notation has been made consistent throughout.

R1.8 — Price and equilibrium metrics are not rigorously defined

“Price and ‘equilibrium’ metrics are not rigorously defined... claims faster convergence.”

Section 3.5 now defines the plotted quantity formally as the cumulative geometric mean of executed Rule-T trade prices per replicate, with the cross-run mean trajectory and cross-run standard deviation $\sigma(t)$ over $R = 50$ replicates. Crucially, we have also **narrowed the claim to what the metric measures**: $\sigma(t)$ quantifies *cross-replicate consistency* of the price trajectory, not within-market dispersion at a single moment, and because a cumulative average contracts mechanically over time, comparisons are made only at matched time points. Section 4.2 has been rewritten so that the claim is more *consistent* (cross-seed-reliable) price convergence, attributed to encounter frequency—the only channel through which movement can affect prices when Rule T is identical across conditions. Within-step dispersion and the Foley (1994) benchmark are identified as future work.

Response to Reviewer #2

We thank Reviewer 2 for a set of comments that materially improved the empirical rigor of the paper, particularly on survival analysis and inequality measurement.

R2.1 — Single-agent PPO with full parameter sharing; recommend reviewing de Witt et al. and an IPPO benchmark

“The most significant methodological limitation is the use of single-agent PPO with full parameter sharing... I recommend the authors review de Witt et al. A useful benchmark would be IPPO.”

We have thought carefully about this recommendation, and we have both (i) added a substantive defense of the parameter-sharing design and (ii) directly tested the underlying question the reviewer raises, through a new ablation experiment.

First, on the design itself, we now argue (Section 3.3.1) that parameter sharing is a deliberate choice aligned with the structure of the canonical *Sugarscape* model rather than a mere computational convenience. In the original model, heterogeneous agents all follow the *same* behavioural rule (Rule M); heterogeneity enters not through different rules but through agents occupying different states (locations, endowments, metabolisms, neighbourhoods). Our shared learned policy preserves exactly this structure: all agents share one decision mapping $\pi_\theta(a \mid o_i)$, yet because their observations differ, their realized behaviour differs. Parameter sharing thus replaces the hand-crafted common rule with a *learned* common rule, while heterogeneity continues to enter through individual states—faithful to the ABM tradition the model belongs to.

Second, we agree with the reviewer’s deeper point that an IPPO design, by letting each agent learn its own policy π_{θ_i} , would relax this homogeneity assumption and could introduce behavioural heterogeneity through the learned rules themselves. We now acknowledge explicitly (Section 5.2) that this is a genuine trade-off and that policy-level heterogeneity could, in principle, give rise to distributional dynamics a shared policy cannot express—an economically interesting question in its own right. We regard this, however, as a *distinct* research question (how heterogeneity in learned rules shapes outcomes), not a robustness check on the present design.

Third, and most directly, the paper’s central question—whether adding market information to movement changes the emergent outcomes—is now tested head-on by a new **ablation of the MRS observation channel** (Section 3.7, the new ablation figure in the revised manuscript). Holding the architecture, hyperparameters, and seed fixed and removing only the MRS channel, we find that the ablated policy fails to learn an effective strategy (cumulative reward stagnates near zero; policy entropy plateaus), whereas the full trade-aware policy learns a decisive, high-reward strategy. This isolates the marginal contribution of market information while keeping the learning architecture and behavioural assumption unchanged—which we believe is a cleaner test of the paper’s claim than an IPPO comparison, since IPPO would vary the architecture and thereby confound the information effect with rule heterogeneity.

For these reasons we have retained the shared-policy design and prioritized the MRS ablation, while citing de Witt et al. (2020) and discussing IPPO as future work. We would add one clarification on scope: because independent per-agent policies would introduce a source of behavioural heterogeneity that is absent from the canonical *Sugarscape* specification, an IPPO formulation constitutes a different model rather than a robustness check on the present one. It would answer a related but separate question—how heterogeneity in the learned rules themselves shapes emergent outcomes—which we see as a valuable direction for subsequent work rather than a modification of the model this paper sets out to study.

R2.2a — Single training run with fixed seeds; report results across ≥ 5 independent training seeds

“The paper trains a single policy with fixed random seeds and reports variance only across evaluation episodes. This conflates environment stochasticity with training stability. . . report results across at least 5 independent training runs with different seeds.”

We have carried out the multi-seed training the reviewer requested. We retrained the policy from scratch under five independent training seeds ($\{1, 2, 3, 42, 123\}$), holding architecture, hyperparameters, and environment fixed, and we now report the learning diagnostics for all five runs (Section 3.7, the new multi-seed figure in the revised manuscript). The five seeds exhibit closely consistent convergence: policy entropy declines along nearly identical trajectories, and cumulative reward rises to a comparable converged range for every seed, with the runs interleaving and no seed emerging as an outlier. This directly addresses the reviewer’s concern that a single run might reflect a fortunate seed: the convergence behaviour is reproducible across independent seeds, not an artifact of one initialization.

We have also clarified the distinction the reviewer rightly drew. The 50 replicates underlying our reported outcomes vary the *environment* seed (resource layout and agent draws) for a converged policy, and therefore establish generalization across layouts; the five-seed experiment separately establishes reproducibility of the *training* process. Section 3.6 now states this distinction explicitly, and the earlier ambiguous wording has been removed.

We are transparent about the one remaining boundary of this analysis. The multi-seed experiment establishes that training *convergence* is consistent across seeds; it does not yet re-evaluate the full suite of downstream macroeconomic outcomes (carrying capacity, prices, inequality, sustainability) separately for each of the five policies. Because the five trained policies are included in the replication package, this per-seed downstream evaluation is a direct next step, which we note in Section 5.2; we expect the outcome variation to be modest given the consistency of the training dynamics, but we refrain from claiming this until it is measured.

R2.2b — The convergence claim looks overstated given $CV > 140\%$

“A mean cumulative reward of 1.05 with SD of 1.48 implies a coefficient of variation exceeding 140%. . . the claim of smooth convergence appears overstated.”

Section 3.7 now addresses the reviewer’s arithmetic directly. We acknowledge that the per-summary reward SD (1.48) is large relative to its mean (1.05), and we explain that this dispersion is expected rather than a sign of training instability: the reward is a per-agent welfare increment aggregated over a heterogeneous population of 500 agents with differing metabolisms, endowments, and local resource access, so individual-step rewards span a wide range even after convergence. We have accordingly rested the convergence assessment not on per-step reward variance but on the rolling-mean reward *trend* (slope below 5×10^{-6}) and the monotonic decline and stabilization of policy entropy, and we have softened the surrounding language.

R2.3 — Survival analysis (Kaplan–Meier, log-rank, optionally Cox) would be more rigorous

*“The comparison of carrying capacity via paired *t*-tests on final population counts discards the rich temporal information. . . Survival analysis, specifically (i) Kaplan–Meier estimation with log-rank testing, and (ii) potentially a Cox proportional hazards model. . .”*

Section 4.1 now includes a Kaplan–Meier survival analysis with log-rank testing (Table 2), reconstructing individual agent lifetimes from the per-step records and treating agents alive at $T = 150$ as right-censored, following standard practice (Kaplan & Meier 1958; Collett 2015).

KM median survival is 45% and 44% longer under DRL on deterministic and stochastic maps respectively, and log-rank tests reject equality of the survival distributions in both settings, confirming that the DRL advantage is present across the full mortality trajectory rather than concentrated at the terminal step. The reviewer’s suggestion (ii) of a Cox proportional-hazards model with agent-level covariates is, as the reviewer notes, optional; we have not included it in this revision but agree it is a natural extension and would be glad to add it if the editor considers it valuable.

R2.4a — Gini is insensitive in fat-tailed distributions; use Theil’s T or top wealth shares

“The paper relies exclusively on the Gini coefficient... fat-tailed, right-skewed wealth distributions are precisely where Gini is least sensitive. Theil’s T or top wealth shares would be more appropriate.”

Section 4.5 now supplements the Gini with two tail-sensitive measures—Theil’s T index and the top-10% wealth share—computed from individual agent wealth at termination (Table 3), citing Atkinson (1970) on the Gini’s insensitivity in this regime. These measures reveal a genuine nuance the Gini alone concealed: on deterministic maps the measures agree that DRL modestly compresses extreme fortunes, but on stochastic maps DRL exhibits significantly *higher* Theil’s T, mechanically traced to the larger share of high-metabolism survivors it sustains. We report this efficiency–equality trade-off transparently and propagate it consistently to the Section 4 summary, the conclusion, and the abstract, without overclaiming in either direction.

R2.4b — Is the Gini confounded by poor agents dying first?

“There is also an additional question: is the Gini coefficient confounded by the fact that poor agents die first during episodes? This needs clarification.”

Section 4.5 now includes a “note on survivorship” that addresses this directly. We acknowledge that all inequality measures are computed over surviving agents and therefore describe the surviving population rather than the full birth cohort, and that because the poorest agents are the most likely to die early, the Gini of survivors mechanically understates cohort-level inequality in both conditions. We further reason through the *direction* of the confound across conditions: because DRL sustains a substantially larger surviving population, its Gini is filtered *less* by early death than the rule-based Gini, so the near-identical terminal Gini values are a conservative comparison—a like-for-like cohort-level comparison would tend to favour DRL rather than undermine it. We flag a full cohort-level treatment (assigning terminal wealth zero to agents that died and recomputing over all agents ever alive) as a refinement for future work, and note that the per-step records required for it are included in the replication package.

R2.5 — Section 5.2 game-AI application claims are unsupported

“Section 5.2 makes application claims to game AI and interactive simulations that are not supported... This subsection should be removed or condensed into a single cautious sentence.”

The standalone game-design subsection has been removed. Its content is now condensed into a single cautious sentence within Section 5.1, which states explicitly that the framework operates in a minimal two-resource environment with no learned trading decisions and that substantial architectural extensions would be required before such applications could be pursued—precisely the gap the reviewer asked us to acknowledge.

Summary of new material added in this revision

For the editor’s convenience, the principal additions are: a Scope-of-Learning paragraph (§3.2); a Dec-POMDP formulation with explicit non-stationarity and a design-level defense of parameter sharing (§3.3.1); a fully specified observation space and reward function (§3.2.1, §3.2.3); a formally defined price metric (§3.5); a single-column parameter table with an identity-of-conditions statement (§3.4, §3.1); full statistical reporting with confidence intervals and effect sizes throughout §4; a Kaplan–Meier survival analysis (§4.1, Table 2); Theil’s T, top-10% share, KS test, and tail-quantile analyses (§4.5, Tables 3–4); a survivorship note on the Gini (§4.5); a multi-seed training-robustness experiment across five independent seeds (§3.7); an ablation of the MRS observation channel (§3.7); and an expanded Limitations section (§5.2). The one item we explicitly scope as future work is the per-seed evaluation of downstream macroeconomic outcomes, which the multi-seed and ablation experiments (both at the level of training dynamics) do not yet cover.

Nine references have been added accordingly: Oliehoek & Amato (2016); Lowe et al. (2017); Foerster et al. (2018); Yu et al. (2022); Kaplan & Meier (1958); Collett (2015); Atkinson (1970); de Witt et al. (2020); and Henderson et al. (2018).

We are grateful to both reviewers; the paper is considerably stronger as a result of their comments.